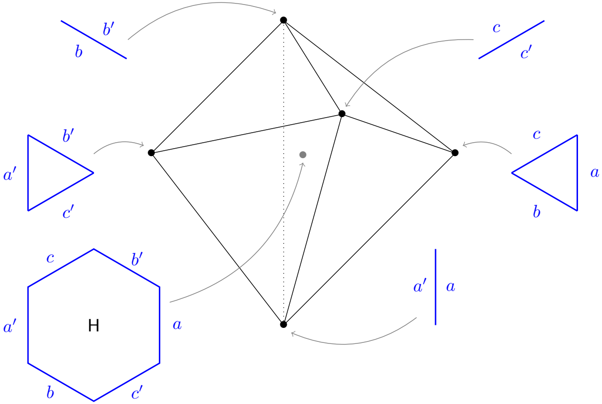
Minkowski indecomposability of polytopes and framework

Germain Poullot with Arnau Padrol

ArXiv: 2512.05307

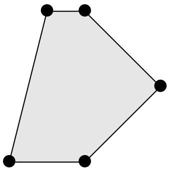


1. What are polytopes and frameworks?
   - Two definitions of polytopes
   - Faces and graphs
2. Deformations (weak Minkowski summands)
   - Minkowski decomposition
   - Cone of deformations
3. Graph of edge dependencies
   - Triangle, parallelograms
   - Graph of edge dependencies
   - Pinning closure and our strongest theorem
4. Applications
   - First applications
   - New indecomposable generalized permutahedra
   - 0/1-polytopes
   - Platonic, Archimedean, and Jhonson solids
   - Products, parallelogramic Minkowski sums
   - More...

What are polytopes and frameworks?

Definition

Polytope: convex hull of finitely many points in $\mathbb{R}^n$



Polytopes

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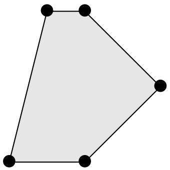
Polytope: convex hull of finitely many points in $\mathbb{R}^n$
bounded intersection of finitly many half-spaces in $\mathbb{R}^n$



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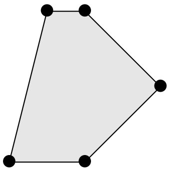

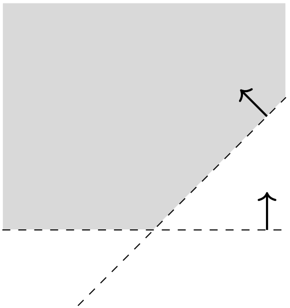

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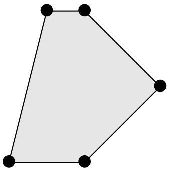

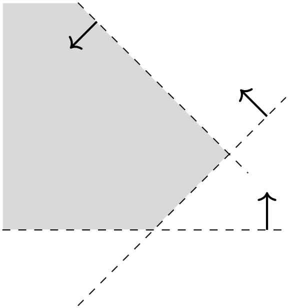

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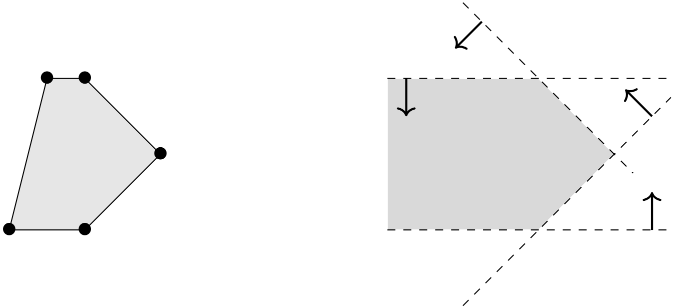

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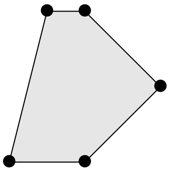

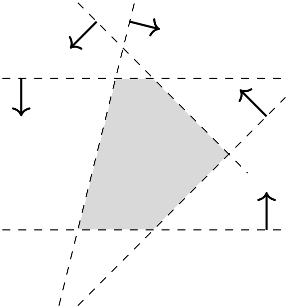

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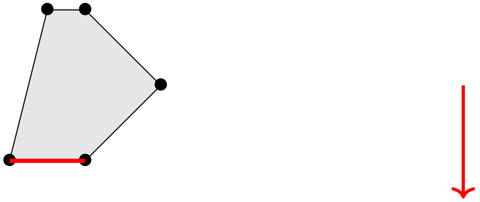

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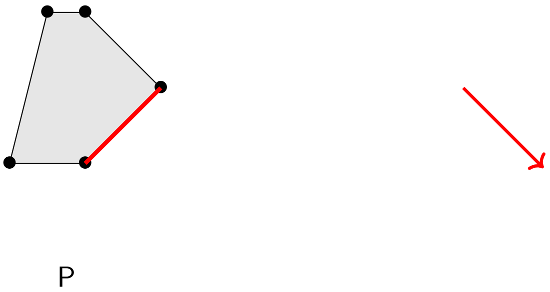

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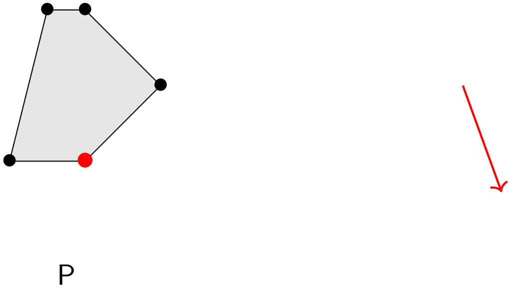
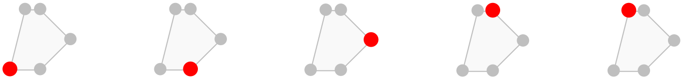


Face lattice

Face lattice: poset of inclusions of faces



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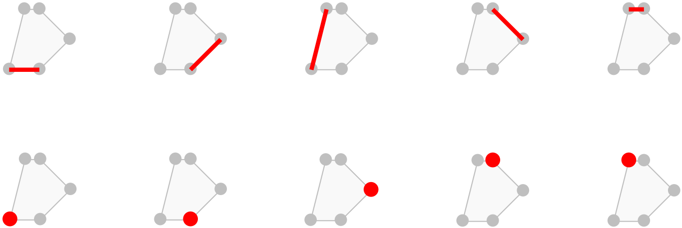

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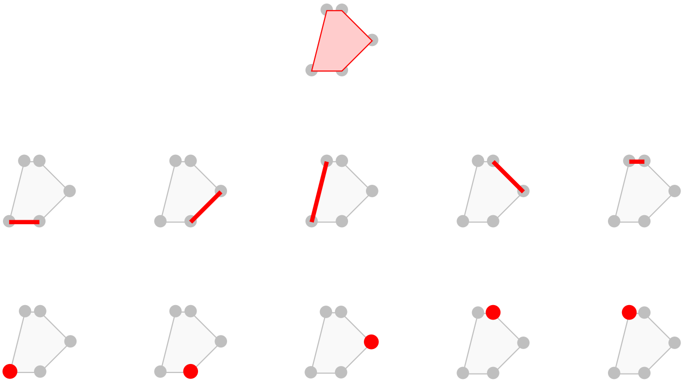

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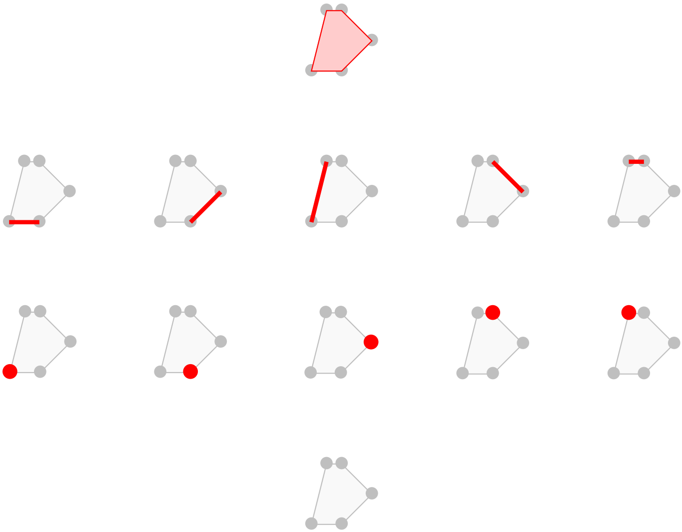

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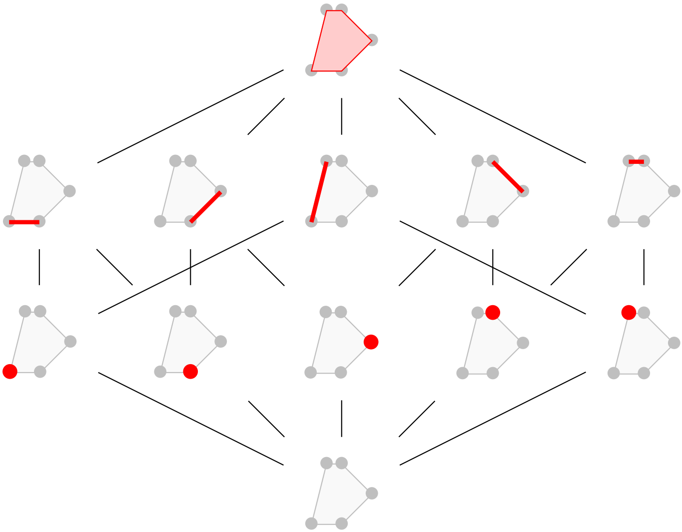

Graph of polytopes *versus* framework

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Framework $\mathcal{F} = (V, E, \phi)$ where (V, E) is a graph, and $\phi : V \to \mathbb{R}^d$ its *realization*.

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Lemma

The cycle space of a polytopal framing is generated by 2-dimensional cycles.

Deformations (weak Minkowski summands)

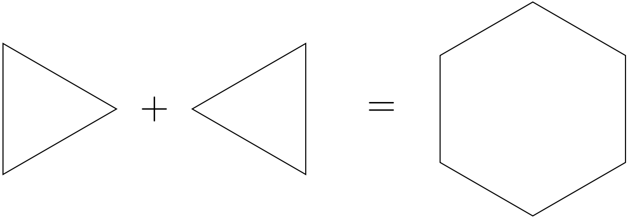
Minkowski sum

Definition

P, Q polytopes. *Minkowski sum*:

$$\mathrm{P} + \mathrm{Q} = \{\boldsymbol{p} + \boldsymbol{q} \;\; ; \;\; \boldsymbol{p} \in \mathrm{P}, \;\; \boldsymbol{q} \in \mathrm{Q}\}$$

N.B. $\mathrm{Vert}(\mathrm{P} + \mathrm{Q}) \subseteq \mathrm{Vert}(\mathrm{P}) + \mathrm{Vert}(\mathrm{Q})$



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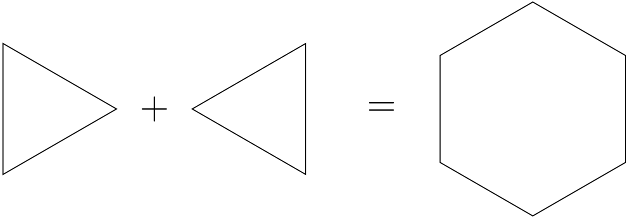
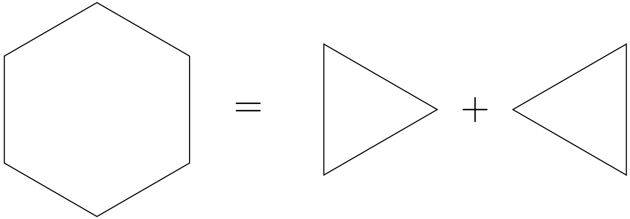
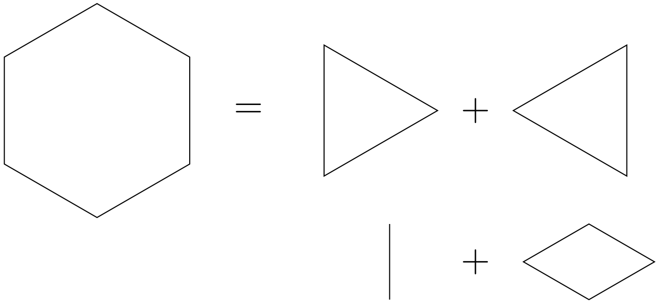


Sum of frameworks $\mathcal{F} + \mathcal{F}'$ with same graph has realization $\phi + \phi'$.

Minkowski sum





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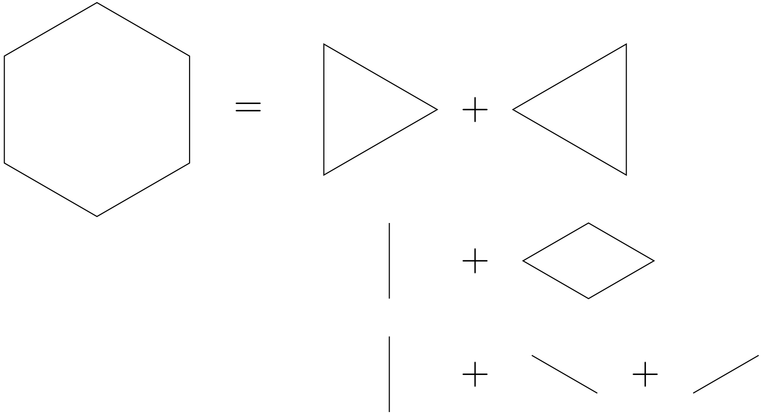

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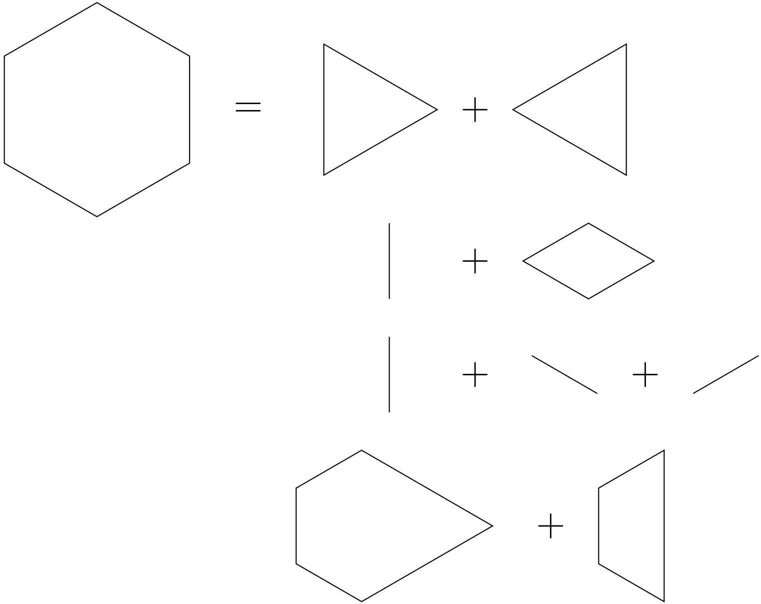

Minkowski summands

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Q is a *Minkowski summand*, a.k.a. *deformation*, of P when there exists R and $\lambda > 0$ with:

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What is the best way to write P as a Minkowski sum ?

- With the fewest number of (indecomposable) summands ?
- With the (indecomposable) summands of smallest dimension ?
- Respecting some symmetries ?
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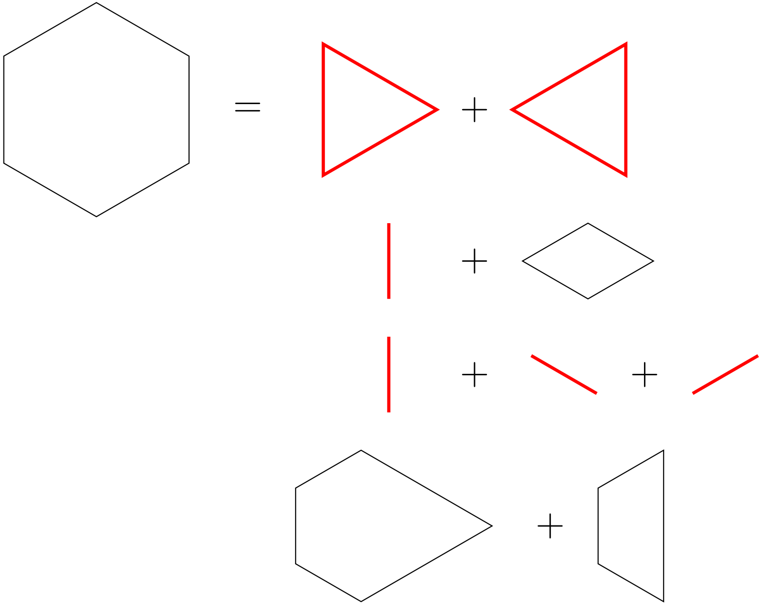
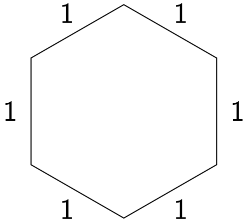
$\Longrightarrow$ What is the structure of $\mathbb{DC}(\mathrm{P})$?

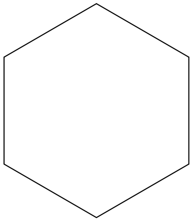
Minkowski summands



Observation

If $P = Q + R$, then the edges of P "are" edges of Q or of R.
$\Rightarrow$ I can write deformations of P using edge-length vectors.





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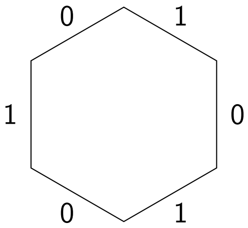

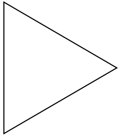

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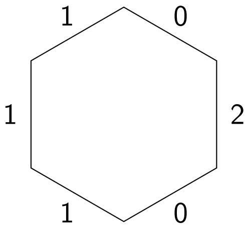

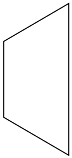

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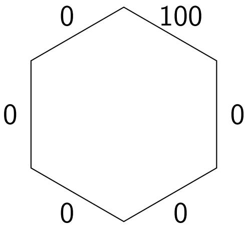

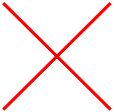

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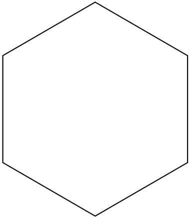
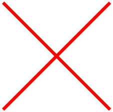
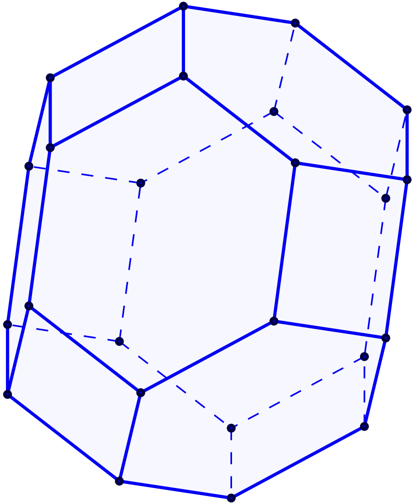


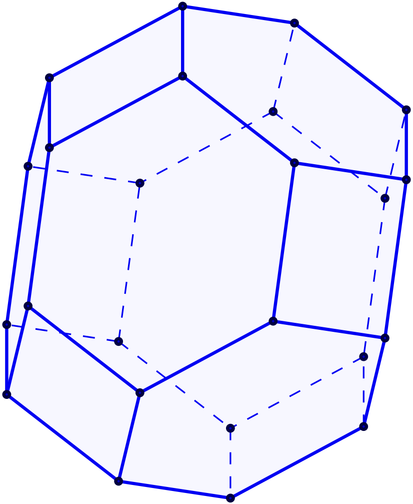


Remark:
As P, Q, R have "same edges", we also have $\mathcal{F}(P) = \mathcal{F}(Q) + \mathcal{F}(R)$.

Deformations of 3-dim permutahedron



Permutahedron Π_4



Sequence of deformations of Π_4

Edge-length deformation cone

Theorem

Q *deformation of* P $\Leftrightarrow$ $\mathcal{F}(\mathrm{Q})$ *deformation of* $\mathcal{F}(\mathrm{P})$

$\Leftrightarrow$ *same edge-directions, but different lengths*

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Deformation cone: $\mathbb{DC}(\mathcal{F}) = \{\mathcal{G}\ ;\ \mathcal{G}$ same graph and edge-dir $\mathcal{F}\}$

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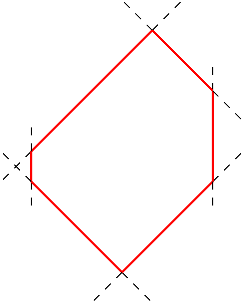

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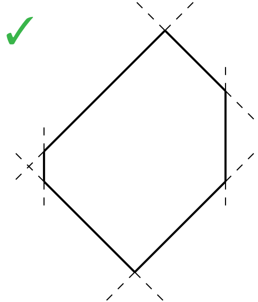

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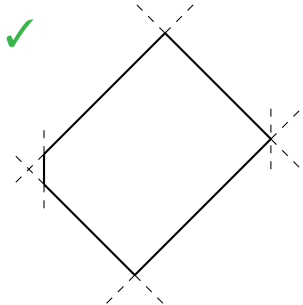

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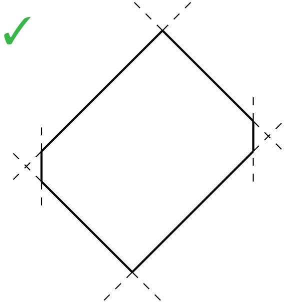

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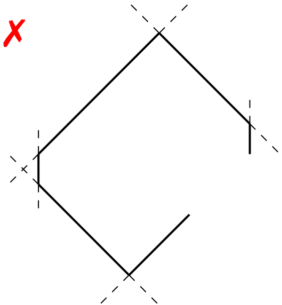

Parametrization:

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Cycle equations:
linear equations on $\boldsymbol{\ell}$
$\ell_e \geq 0$ for all e edge

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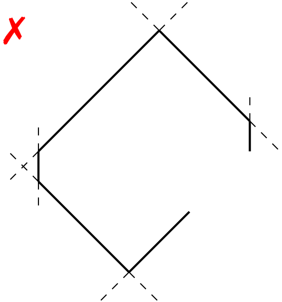

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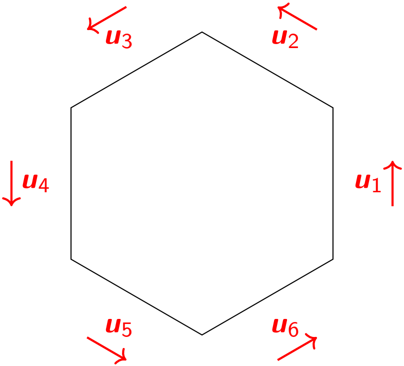

For a cycle in $\mathcal{F}$:

$$\sum_{e \in \text{cycle}} \boldsymbol{u}_e = \mathbf{0} \quad , \quad \boldsymbol{u}_e \text{ edge vector}$$

hence

$$\sum_{e} \ell_e \boldsymbol{u}_e = \mathbf{0}$$

$\mathbb{DC}(\mathrm{P})$

Q	$\boldsymbol{\ell}$	$\boldsymbol{h}$
Minkowski summands	edge-lengths	heights on rays
$\mathrm{Q}_1 + \mathrm{Q}_2$	$\boldsymbol{\ell}_1 + \boldsymbol{\ell}_2$	$\boldsymbol{h}_1 + \boldsymbol{h}_2$
Dilation $\lambda\mathrm{Q}$	$\lambda\boldsymbol{\ell}$	$\lambda\boldsymbol{h}$
Translations	Has been fixed	Lineal
	edge directions	normal fan $\mathcal{N}_\mathrm{P}$
complicated	Cycle equations	Wall-crossing ineq.
	V-description	H-description
Polytope algebra	Weight algebra	Polynomial algebra

$\mathbb{DC}(\mathrm{P})$ is a ray = P Minkowski indecomposable

$\mathbb{DC}(\mathrm{P})$ is simplicial cone = P has **unique** Minkowski decomposition

Graph of edge dependencies

Cycle equations of a triangle, quadrilateral

Goal: Solve quickly $\{\text{cycle equations}\} \cap \{\text{positive variables}\}$.

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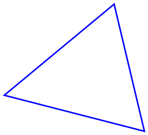

Triangle:
3 variables (= lengths of the edges)
2 equations (= 1 equation in dimension 2)
$\Rightarrow$ dim(space of solutions) $= 1$

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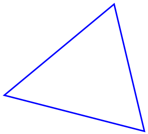

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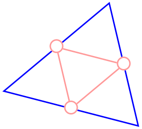

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$\Rightarrow$ If I know the length of 1 edge, I know the length of the others.

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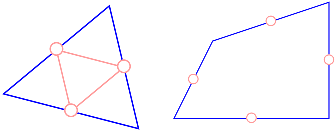

Triangle:

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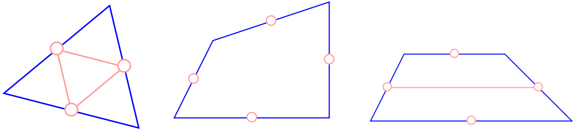

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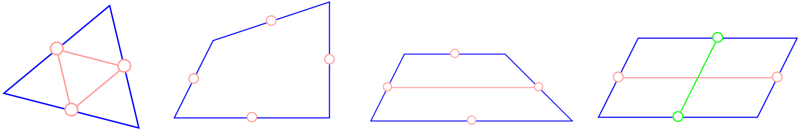

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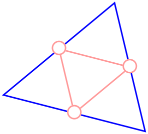
Graph of edge dependencies

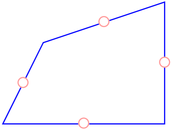
Graph of edge dependencies $ED(\mathcal{F})$:

nodes: edges of framework $\mathcal{F}$

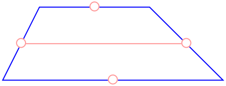
arcs: link two *dependent* edges, i.e. I can deduce the length of one from the length of the other, using the cycle equations



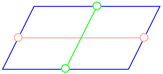
(a) Triangle



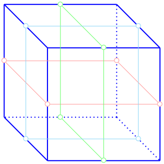
(b) Quadrilateral



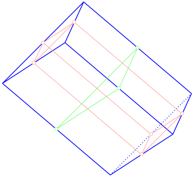
(c) Trapezoid



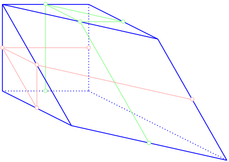
(d) Parallelogram



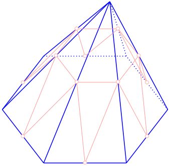
(e) Cube



(f) Prism



(g) Hemicube



(h) Pyramid

Problem 1:
How to find arcs of $ED(\mathcal{F})$.

The two problems

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How to find arcs of $ED(\mathcal{F})$.

Problem 2:
If I have enough arcs in $ED(\mathcal{F})$, what can I conclude on $\mathbb{DC}(\mathcal{F})$ or on the indecomposability of $\mathcal{F}$?

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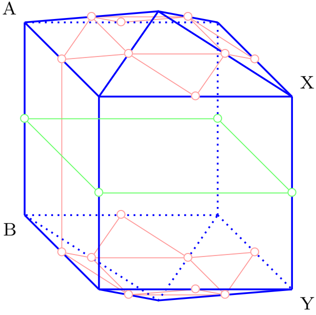
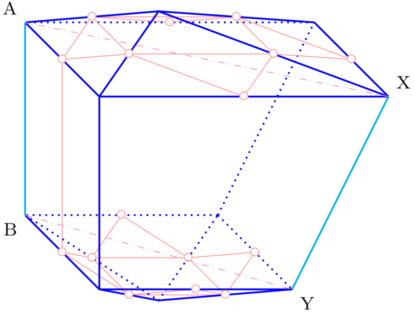
"triangles" $\rightarrow$ arcs in $ED(\mathcal{F}(\mathrm{P}))$
"all 2-faces" $\rightarrow$ enough arcs in $ED(\mathcal{F}(\mathrm{P}))$

Problem 1: creating edges

In $ED(\mathcal{F})$, two nodes e, f ($\in$ edges of $\mathcal{F}$) are linked if, e.g.:

- in a common triangle (not necessarily a 2-face),





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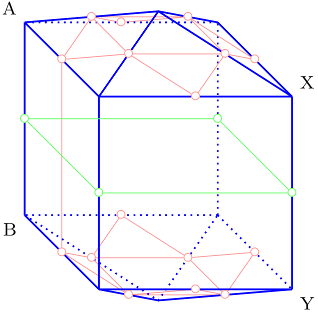

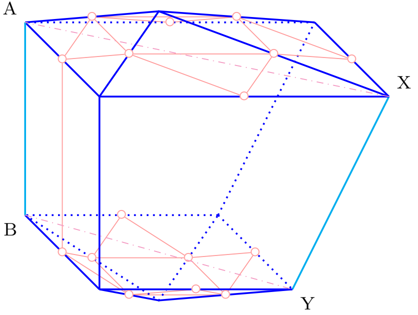

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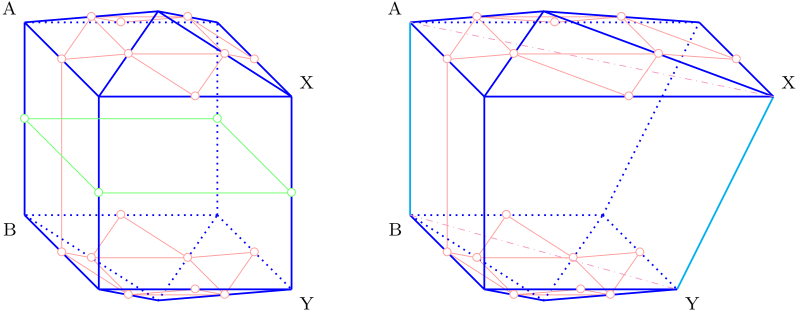

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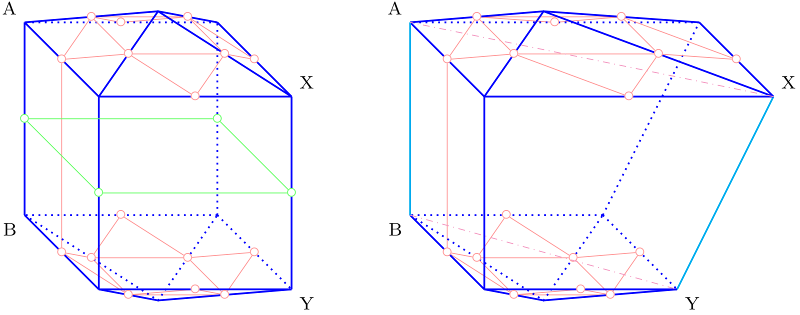

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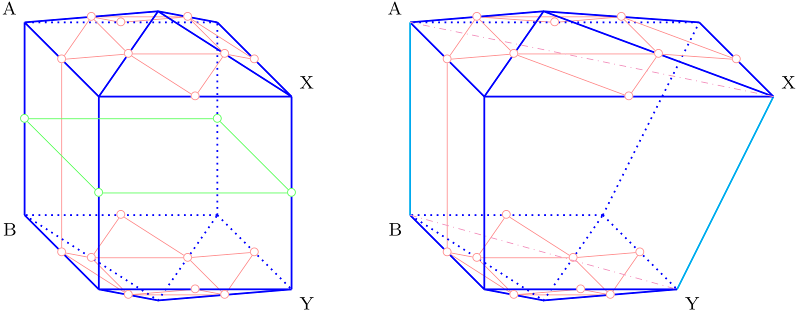

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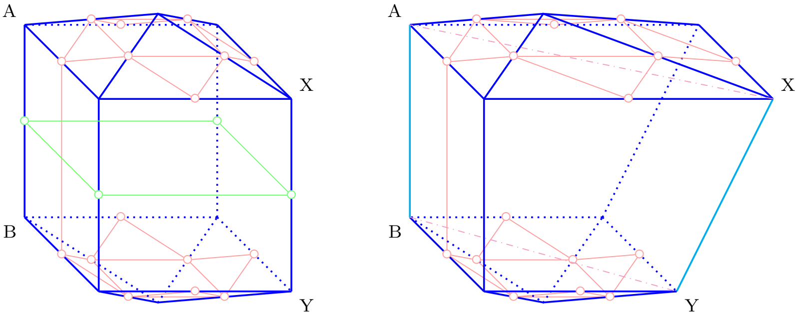
- ... (there is a nicer way to write these properties)

+ implicit edges



Problem 2: consequences of $ED(\mathcal{F})$

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Dependent $S \subseteq V$: for all $u, v \in S$, there is a path (in graph of $\mathcal{F}$) of dependent edges between u and v.

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Corollaries: Gale '54, Shepard '63, Kallay '82, McMullen '87, Yost–Przesławski '08 & '16 criteria…

Pinning closure

flat: set of vertices inducing a *connected* sub-framework of $\mathcal{F}$
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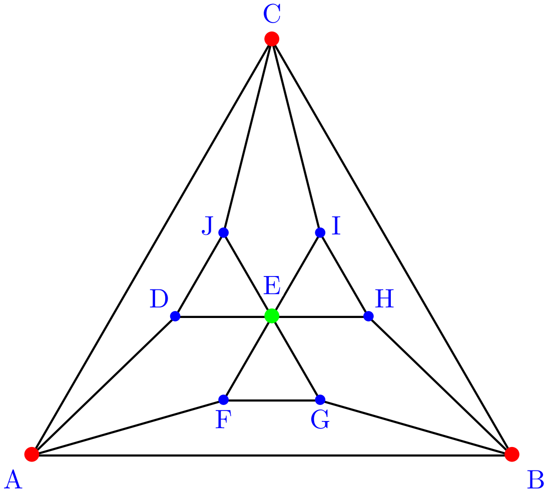
pinning closure: $\mathrm{cl}(S) = max_i S_i$

Pinning closure



This is indecomposable

Our strongest theorem

Theorem (Padrol–P. '25+)

If there is $S \subseteq V$, and $X_1, \dots, X_r$ dependent sets of edges with

- *any two vertices of S are connected via $\bigcup_i X_i$, and*
- $\mathrm{cl}(S) = V$,

then: $\dim \mathbb{DC}(\mathcal{F}) \leq r$.

Corollary (Padrol–P. '25+)

If there is S such that:

- *S is a dependent set of vertices*
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then: $\mathcal{F}$ is indecomposable.

Applications

First applications

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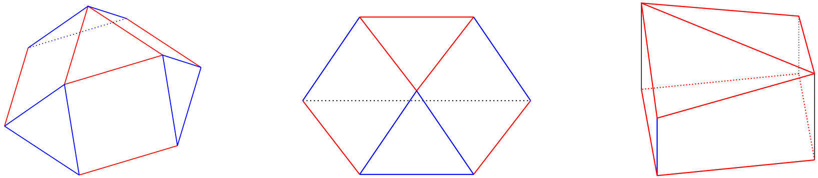
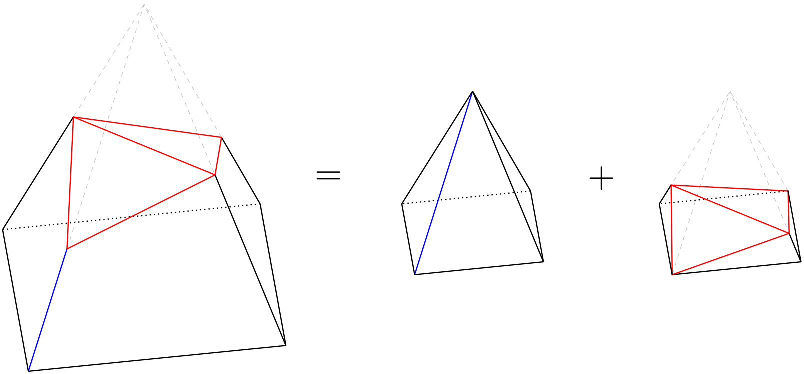


(a) Triang. cupola (b) Sum of 2 triangles (c) Chiseled cube

$\dim \mathbb{DC}(\mathrm{P}) = 2$, only one Minkowski decomposition (in 2 terms)

First applications



$\dim \mathbb{DC}(P) = 2$, this is the only Minkowski decomposition

New indecomposable generalized permutahedra

Generalized permutahedra: edge directions are $\boldsymbol{e}_i - \boldsymbol{e}_j$ for $i \neq j$
Edmonds problem '70: find all indecomposable gene. permut.

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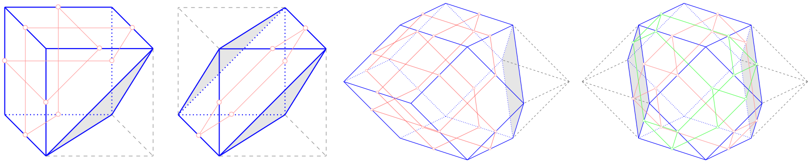
Theorem (Padrol–P. '25+)

For all complete bipartite graphs $K_{n,m}$ (except $n = m = 2$), it is possible to truncate 1 or 2 vertices of $\mathrm{Z}_{K_{n,m}}$ to obtain an indecomposable generalized permutahedron.



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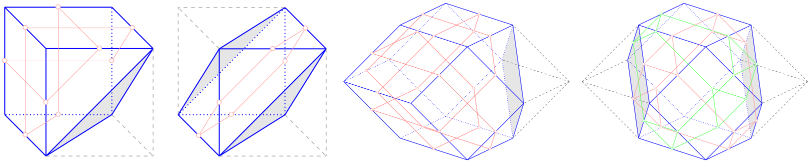

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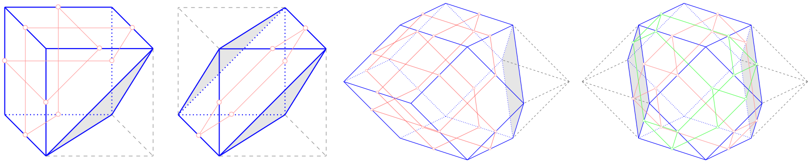

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Loho–Padrol–P.'25: we create $2^{2^{n-2}}$ indecomposable gene. permut.

0/1-polytopes

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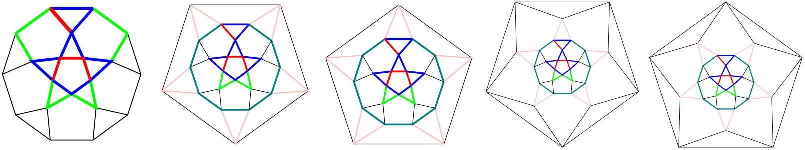
Easy proof: matroid pol. indecomposable $\Leftrightarrow$ matroid connected

See Higashitani–Padrol–Sanyal for more (soon to come).

Platonic solids: 3-dim., 1 type of regular polygons, vertex transitive
Archimedean solids: 3-dim., regular polygons, vertex transitive
Jhonson solids: 3-dim., regular polygons



(a) J_6 (b) J_{32} (c) J_{33} (d) J_{34} (e) A_8

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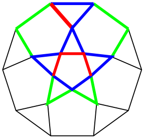
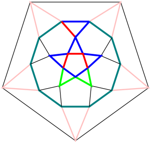
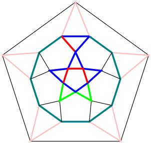
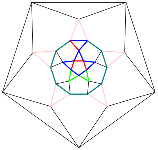
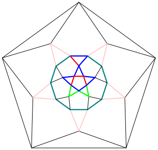

(a) J_6


(b) J_{32}


(c) J_{33}


(d) J_{34}


(e) A_8

Theorem (P. '26+)

$\dim \mathbb{DC}(P)$ *for all Platonic, Archimedean, and Jhonson solids: 2 infinite families + 110 sporadic solids*

Products, parallelogramic Minkowski sums

Theorem (Padrol–P. '25+)

$\mathbb{DC}(\mathsf{P} \times \mathsf{Q}) \simeq \mathbb{DC}(\mathsf{P}) \times \mathbb{DC}(\mathsf{Q})$

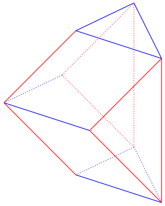
Theorem (Padrol–P. '25+)

$\mathbb{DC}(P \times Q) \simeq \mathbb{DC}(P) \times \mathbb{DC}(Q)$

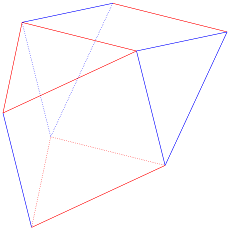
P, Q are in *parallelogramic position* if no edge of P is in a plane spanned by the 2-faces of Q

Theorem (Padrol–P. '25+)

P, Q *in parallelogramic position:* $\mathbb{DC}(P + Q) \simeq \mathbb{DC}(P) \times \mathbb{DC}(Q)$.



(a) Diminished trapezohedron



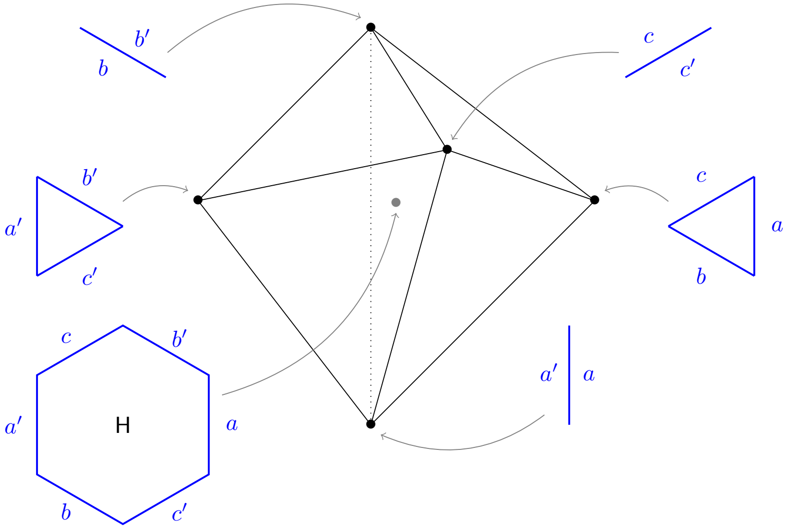
(b) Gyrobifastigium

Other applications:

- "autonomous & dependent" parts of $ED(\mathcal{F})$ give rays of $\mathbb{DC}(\mathcal{F})$
- edge decomposability implies unique decomposability property
- stack/delete vertices on polytopes, then study indecomposability
- you can deal with zonotopes whose 2-faces are parallelograms
- accelerate the computation $\mathbb{DC}(\mathcal{F})$ using combinatorial data
- ...

Remark:

Be careful, $ED(\mathcal{F})$ is not always sufficient for recovering $\mathbb{DC}(\mathcal{F})$.



Merci !   Thank you!

Bonus slides...

What else do I do?

Submodular cone: "Many rays of the submodular cone"

Linear programming: "Vertices of the monotone path polytopes of hypersimplicies" & "Pivot polytopes of products of simplices and shuffles of associahedra"

Random polytopes: "Unimodality of the number of paths per length on polytopes: Examples, counter-examples, and central limit theorem"

Ehrhart theory, triangulations: "Ehrhart non-positivity and unimodular triangulations for classes of s-lecture hall simplices"

3D polytopes, framing lattices, polytope/weight algebra, hypergraphs, Gray codes, hypergraphs...

Anything that might contain a polytope somewhere is interesting !!